

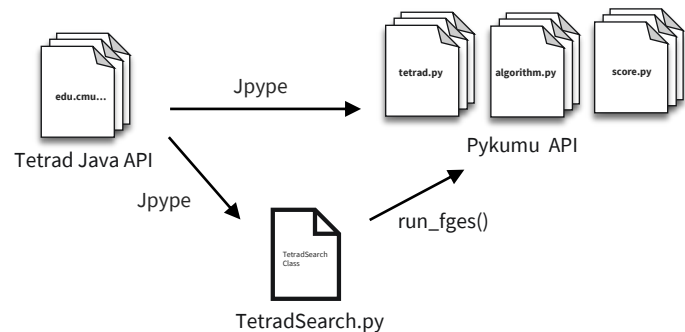
# Pykumu :: CHEAT SHEET



## About

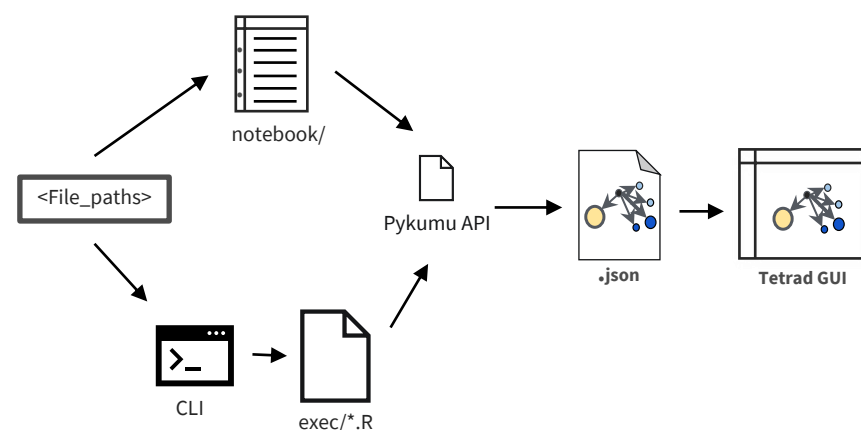
Pykumu is a Python package that serves as a wrapper over the Tetrad Java API (via JPytype), providing an interface for causal analysis algorithms. Pykumu is a fork of py-tetrad/pytetrad.

## Pykumu API



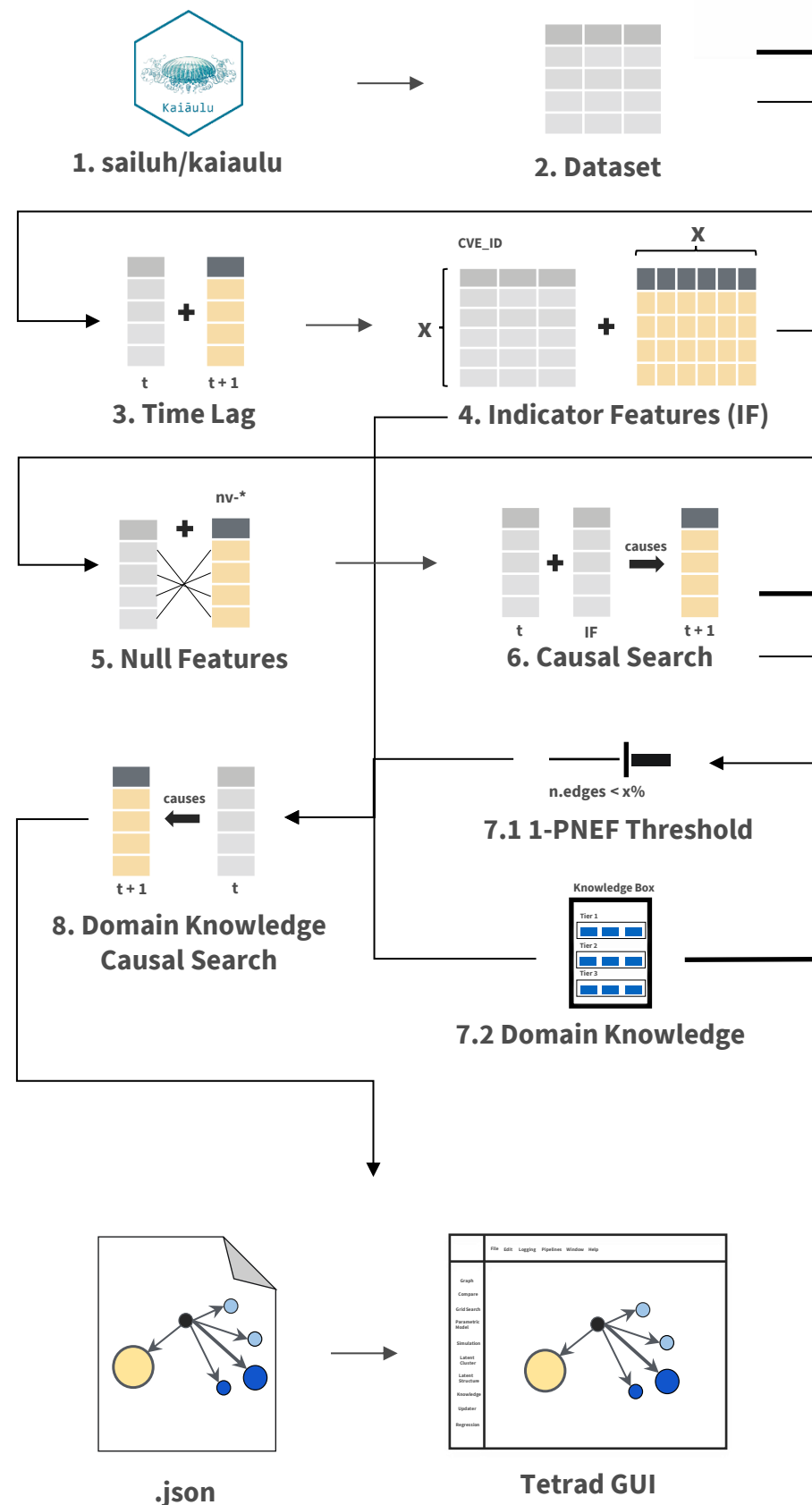
Pykumu wraps the Tetrad Java API through JPytype, exposing individual modules in pykumu/api (data, score, algorithm, graph, knowledge, bootstrapping) that provide direct function level control over each step of the causal search pipeline. Pykumu uses an R Package architecture, like Kaiaulu, with API, Notebooks, and Scripts. Its scripts and Notebooks also support interfaces to other tools in the Sailuh tools ecosystem.

## Command Line Interface



Exec scripts provides command line access to Pykumu API package, allowing its causal analysis capabilities to be accessed via the terminal or server side without requiring Python code knowledge.

## Causal Analysis Workflow



## API

Pykumu more closely interfaces with Kaiaulu but can use any other dataset. For an example of a dataset, see **issue\_social\_smell\_showcase.Rmd**. Refer to [cheatsheets/social-smells-cheatsheet.pdf](#)

### tetrad.py

After applying time lag and features, jpytype is imported to utilize API and turn the dataset into a Tetrad compatible format.

#### 1. tetrad\_jvm\_start()

Import jpytype to start JVM and load Tetrad jar file.

#### 2. transform\_pandasdf\_to\_tetrad\_boxdataset()

Convert pandas DataFrames into Tetrad BoxDataset and initialize search state.

### causal.py

Causal search is run on dataset including null features, used to derive 1-PNEF.

#### 1. score\_sem\_bic()

Configure SEM BIC score.

#### 2. bootstrapping()

Configure bootstrapping parameters.

#### 3. algorithm\_fges()

Performs Fast Greedy Equivalence Search (FGES) algorithm.

#### 4. algorithm\_boss()

Performs Best Order Score Search (BOSS) algorithm.

### knowledge.py

Causal search can be used with domain knowledge applied.

#### 1. parse\_knowledge\_txt()

Load forbidden/required edges and temporal tiers from a .txt file into Tetrad knowledge object.

### graph.py

Parsing graph data into nodes, an edge set, and edge-type probabilities enables 1-PNEF to be applied. Causal search graph outputs can be saved as JSON, which can be loaded and viewed in the Tetrad GUI.

#### 1. transform\_graph\_java\_to\_graph\_json()

Convert Tetrad graph object to JSON format.

#### 2. parse\_graph\_json()

Parse Tetrad JSON data into nodes, edgeset, and edge\_type\_probabilities datasets.